

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 17

Open Questions and the Shape of the Unknown

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. The 13 key ideas from the curriculum, in my own words

(write each one: this is the final consolidation)

- a.
- b.
- c.
- d.
- e.
- f.
- g.
- h.
- i.
- j.
- k.
- l.
- m.

2. The vocabulary I now have that I did not have before

list at least 12 terms with one-sentence definitions in your own words, include at least one each from dynamics, probability/universality, measurement, and computation

3. For each open question listed in the lecture, my current

position:

- a. Why is the universe compressible? (and: what does universality explain, and what does it leave open?)
- b. What is the right foundation?
- c. Discovered or invented?
- d. Computational complexity of finding structure?
- e. Why is correct mathematics beautiful?
- f. What is probability?

4. The personal knowledge gaps I most want to fill next

rank the list from the lecture by priority, with one sentence explaining why for each

5. The original questions I came in with

- a. What is mathematics? → my current answer:
- b. Why does it work? → my current answer:
- c. What are numbers? → my current answer:
- d. How do the fields relate? → my current answer:
- e. What happens when a model works? → my current answer:

6. What I still don't know but can now name

this is the most important section, what shape does the unknown have now?
